

Pandas

- used for data manipulation, data analysis (data inspection) and data cleaning.
- It helps us to work with structured data in the form of rows and columns. mainly used to read, organize, clean, analyze and manipulate the data.
- Data cleaning
 - Removing ~~the~~ null values
 - Remove duplicates
 - replace incorrect values
 - handle missing data
- Data Analysis
 - average
 - maximum
 - minimum
 - Sum
 - grouping
 - sorting
 - filtering
- Data Visualization
 - matplotlib
 - seaborn
- Series
 - one dimensional data structure
 - single column in excel
 - one dimensional labelled array capable of storing any type of data
- dataframe
 - two dimensional data structure
 - excel spreadsheet - rows and columns
 - two dimensional labelled data structure consisting rows and columns

→ Pip install pandas

* Series S capital *

```
import pandas as pd
```

```
marks = pd.Series([10, 20, 30, 40, 50])
```

```
print(marks)
```

```
// 0    10
```

```
1    20
```

```
2    30
```

```
3    40
```

```
4    50
```

```
dtype: int64
```

```
data = {"name": ["john", "david", "sara"],
```

```
       "age": [25, 30, 35]
```

```
       "city": ["chennai", "mumbai", "delhi"]}
```

* DataFrame

DKF Caps*

```
df = pd.DataFrame(data)
```

```
print(df)
```

```
//   name  age  city
```

```
0  john   25  chennai
```

```
1  david  30  mumbai
```

```
2  sara   35  delhi
```